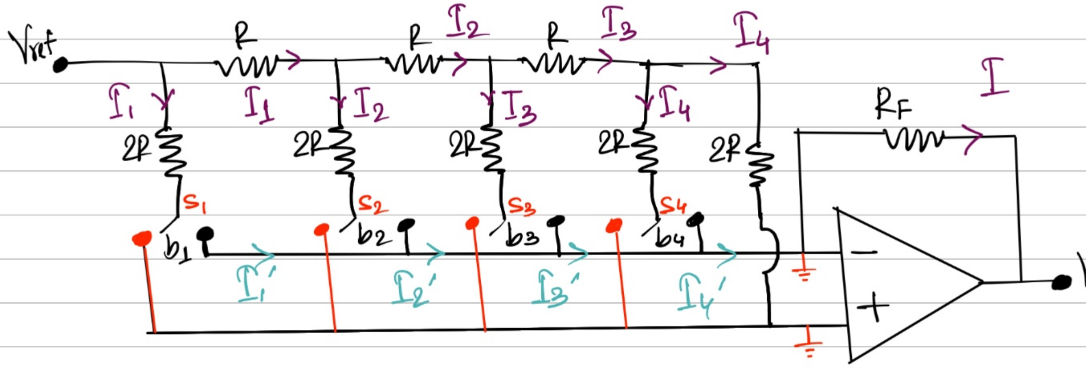
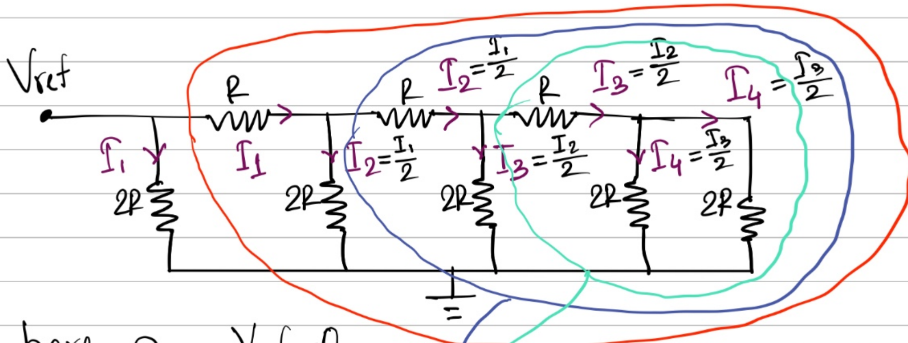
R-2R Ladder DAC

4-bit R-2R DAC





here, $I_1 = \frac{V_{ref} - 0}{2R}$

$\therefore \boxed{I_1 = \frac{V_{ref}}{2R}}$

$\therefore I_2 = \frac{I_1}{2} = \frac{V_{ref}}{4R}$

$I_3 = \frac{I_2}{2} = \frac{V_{ref}}{8R}$

$I_4 = \frac{I_3}{2} = \frac{V_{ref}}{16R}$

$\rightarrow (((2R \| 2R) + R) \| 2R) + R) \| 2R) + R$

$= ((((R+R) \| 2R) + R) \| 2R) + R$

$= (((2R \| 2R) + R) \| 2R) + R$

$= ((R+R) \| 2R) + R$

$= (2R \| 2R) + R$

$= R + R$

$= 2R$

$\rightarrow$ similarly $\rightarrow 2R$

$\rightarrow 2R$

If $b_1 = 1 \rightarrow S_1$ is connected to inverting i/p $\rightarrow \therefore I_1' = I_1$

If $b_1 = 0 \rightarrow S_1$ " " " Non-inverting i/p $\rightarrow I_1' = 0$

$\therefore \boxed{I_1' = b_1 I_1}$, $I_2' = b_2 I_2$, $I_3' = b_3 I_3$, $I_4' = b_4 I_4$

here, $I = \dfrac{0 - V_o}{R_F} = \dfrac{-V_o}{R_F}$

Applying KCL

$$I = I_1' + I_2' + I_3' + I_4'$$

$$\Rightarrow \frac{-V_o}{R_F} = b_1 I_1 + b_2 I_2 + b_3 I_3 + b_4 I_4$$

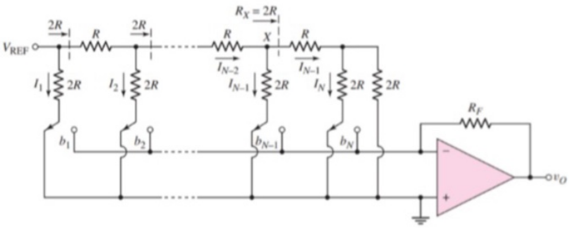
$$\Rightarrow -\frac{V_o}{R_F} = b_1 \frac{V_{ref}}{2R} + b_2 \frac{V_{ref}}{4R} + b_3 \frac{V_{ref}}{8R} + b_4 \frac{V_{ref}}{16R}$$

$$\Rightarrow \boxed{V_o = -V_{ref} \frac{R_F}{2R} \left(b_1 + \frac{b_2}{2} + \frac{b_3}{4} + \frac{b_4}{8} \right)}$$

let 4 bit

- * Step size → 1 LSB value → V_o corresponds to 0001
- * 1 MSB value → V_o corresponds to 1000
- * Max o/p voltage → V_o " " 1111 → V_{max}
- * Min o/p " → V_o " " 0000 → V_{min}

Practice Problem



The N-bit D/A converter with an R–$2R$ ladder network in the above figure is to be designed as a 6-bit D/A device. Suppose $V_{REF} = -5V$ and $R_F = R = 5k\Omega$.

(a)	Find currents $I_1, I_2, I_3, I_4, I_5, I_6$.
(b)	What is the output voltage if the input is 010011?
(c)	What is the change in output voltage if the input changes from 101010 to 010101?

a) $I_1 = \dfrac{V_{REF}}{2R} = \dfrac{-5}{10} \Rightarrow I_1 = -0.50mA$

$I_2 = \dfrac{I_1}{2} = -0.25mA$

$I_3 = \dfrac{I_2}{2} = -0.125mA$

$I_4 = \dfrac{I_3}{2} = -0.0625mA$

$I_5 = \dfrac{I_4}{2} = -0.03125mA$

$I_6 = \dfrac{I_5}{2} = -0.015625 mA$

b) Input $= 010011$

$v_o = -[I_2 + I_5 + I_6] R_F = [0.25 + 0.03125 + 0.015625] mA \times 5k\Omega = \boxed{1.484375V}$

c) Input $= 101010$

$v_o = -[I_1 + I_3 + I_5] R_F = [0.50 + 0.125 + 0.03125] mA \times 5K\Omega = 3.28125V$

Input $= 010101$

$v_o = -[I_2 + I_4 + I_6] R_F = [0.25 + 0.0625 + 0.015625] mA \times 5k\Omega = 1.640625V$

$\boxed{\Delta V_o = 1.640625V}$